

Java Priority Queue:-

Code:-

```
import java.util.ArrayList;
import java.util.List;
import java.util.Scanner;
import java.util.PriorityQueue;

class Student {
    private int id;
    private String name;
    private double cgpa;

    public Student(int id, String name, double cgpa) {
        this.id = id;
        this.name = name;
        this.cgpa = cgpa;
    }

    public int getID() {
        return id;
    }

    public String getName() {
        return name;
    }

    public double getCGPA() {
        return cgpa;
    }
}

class Priorities {
    public List<Student> getStudents(List<String> events) {
        PriorityQueue<Student> queue = new PriorityQueue<>(
            (a, b) -> {
                if (a.getCGPA() != b.getCGPA()) {
                    return Double.compare(b.getCGPA(), a.getCGPA());
                }
                int nameCompare = a.getName().compareTo(b.getName());
            }
        );
    }
}
```

```

        if (nameCompare != 0) {
            return nameCompare;
        }
        return Integer.compare(a.getID(), b.getID());
    }
};

for (String event : events) {
    String[] parts = event.split(" ");
    if (parts[0].equals("ENTER")) {
        String name = parts[1];
        double cgpa = Double.parseDouble(parts[2]);
        int id = Integer.parseInt(parts[3]);
        queue.add(new Student(id, name, cgpa));
    } else if (parts[0].equals("SERVED")) {
        if (!queue.isEmpty()) {
            queue.poll();
        }
    }
}

List<Student> result = new ArrayList<>();
while (!queue.isEmpty()) {
    result.add(queue.poll());
}

return result;
}
}

public class Solution {
    private final static Scanner scan = new Scanner(System.in);
    private final static Priorities priorities = new Priorities();
    public static void main(String[] args) {
        int totalEvents = Integer.parseInt(scan.nextLine());
        List<String> events = new ArrayList<>();
        while (totalEvents-- != 0) {
            String event = scan.nextLine();

```

```

events.add(event);
}
List<Student> students = priorities.getStudents(events);
if (students.isEmpty()) {
    System.out.println("EMPTY");
} else {
    for (Student st: students) {
        System.out.println(st.getName());
    }
}
}
}
}

```

Output:-

Test case 0

Test case 1

Test case 2

Test case 3

Test case 4

Test case 5

Test case 6

Compiler Message

Success

Input (stdin)

1

12

2

ENTER John 3.75 50

3

ENTER Mark 3.8 24

4

ENTER Shafaet 3.7 35

5

SERVED

6

SERVED

7

ENTER Samiha 3.85 36

8

SERVED

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Java ArrayList:-

Code:-

```
import java.io.*;
import java.util.*;
public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        ArrayList<ArrayList<Integer>> list = new ArrayList<>()
        for (int i = 0; i < n; i++) {
            int d = sc.nextInt();
            ArrayList<Integer> row = new ArrayList<>();
            for (int j = 0; j < d; j++) {
                row.add(sc.nextInt());
            }
            list.add(row);
        }
        int q = sc.nextInt();
        for (int i = 0; i < q; i++) {
            int x = sc.nextInt();
            int y = sc.nextInt();
            if (x >= 1 && x <= n && y >= 1 && y <= list.get(x - 1).size()) {
                System.out.println(list.get(x - 1).get(y - 1));
            } else {
                System.out.println("ERROR!");
            }
        }
        sc.close();
    }
}
```

Output:-

✓ **Test case 0**

Compiler Message

✓ Test case 1

Success

✓ Test case 2 

Input (stdin)

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✓ Test case 3 

✓ Test case 4 

✓ Test case 5 

```
1 5
2 5 41 77 74 22 44
3 1 12
4 4 37 34 36 52
5 0
6 3 20 22 33
7 5
8 1 3
```